

CodeAlpha Internship Project Report

Domain:	Artificial Intelligence	Academic Year:	B.Tech 2nd Year
Project Title:	Task 4 - Object Detection and Tracking	Submission Date:	June 2026

1. Project Overview

Yeh project Computer Vision domain par based hai. Isme system ke main webcam ya external video feed ka use karke real-time me objects (jaise humans, laptops, bottles, cars, etc.) ko detect aur track kiya jata hai. Is task ke liye humne industry-standard deep learning model **YOLO (You Only Look Once)** ka use OpenCV package ke saath kiya hai.

2. Implementation Approach

- **Video Capture:** OpenCV library (cv2.VideoCapture) ka use karke camera frame stream receive kiya jata hai.
- **AI Deep Learning Model:** Ultralytics library se pre-trained yolov8n.pt (nano version, lightweight) model automate download aur run hota hai.
- **Bounding Boxes & Visualization:** Model dynamically objects ko locate karke unpar rectangle grids draw karta hai aur labels print karta hai with dynamic tracking coordinates.

3. Python Source Code

```
import cv2
try:
    from ultralytics import YOLO
except ImportError:
    print("Please install ultralytics package: pip install ultralytics")
    exit()

def run_object_detection():
    # Load the pre-trained lightweight YOLOv8 model
    # It automatically detects 80 everyday common objects (person, phone, cup, laptop
    etc.)
    model = YOLO("yolov8n.pt")

    # Initialize real-time webcam feed (0 is standard internal webcam)
    cap = cv2.VideoCapture(0)

    if not cap.isOpened():
        print("Error: Could not open webcam feed.")
        return
```

```

print("Starting Object Detection... Press 'q' key to quit the camera window.")

while True:
    ret, frame = cap.read()
    if not ret:
        print("Failed to read frame from camera. Exiting...")
        break

    # Run YOLO inference on current frame
    # tracking=True applies SORT/ByteTrack algorithms internally for object
tracking IDs
    results = model.track(frame, persist=True, verbose=False)

    # Plot outcomes directly onto the frame
    annotated_frame = results[0].plot()

    # Display the live window
    cv2.imshow("CodeAlpha AI Task 4 - Object Detection & Tracking",
annotated_frame)

    # Break loops if user clicks 'q'
    if cv2.waitKey(1) & 0xFF == ord('q'):
        break

# Cleanup tasks
cap.release()
cv2.destroyAllWindows()

if __name__ == "__main__":
    run_object_detection()

```

4. Prerequisites & Installation

Is advanced Computer Vision module ko apne local machine par run karne ke liye dependencies install karein:

- `pip install opencv-python ultralytics`